# THE DIAGONAL OF THE PICARD FUNCTOR OF A FLAT PROPER FAMILY IS A LOCALLY CLOSED IMMERSION

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 9198-9215 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 9217-9304, the complete author proof (86 body lines).

*Editorial note.* Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The Picard functor and rigidified line bundles, cohomology and base change for flat proper families, and supports of coherent modules; the locally-closedness argument is retained.

*Source reference.* Original lemma cohomology-lemma-projection-formula: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9304–9304. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma morphisms-lemma-fppf-open: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9304–9304. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma perfect-lemma-flat-proper-perfect-direct-image-general: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9304–9304. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma perfect-lemma-locally-closed-where-H0-locally-free: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9304–9304. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let $f : X \to S$ be a flat, proper morphism of finite presentation. Let $\mathcal{E}$ be a finite locally free $\mathcal{O}_X$-module. For a morphism $g : T \to S$ consider the base change diagram*

$$\begin{array}{ccc} X_T & \xrightarrow{q} & X \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle f} \\ T & \xrightarrow{g} & S \end{array}$$

*Date*: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

*Assume $\mathcal{O}_T \to p_*\mathcal{O}_{X_T}$ is an isomorphism for all $g : T \to S$. Then there exists an immersion $j : Z \to S$ of finite presentation such that a morphism $g : T \to S$ factors through $Z$ if and only if there exists a finite locally free $\mathcal{O}_T$-module $\mathcal{N}$ with $p^*\mathcal{N} \cong q^*\mathcal{E}$.*

*Proof.* Observe that the fibres $X_s$ of $f$ are connected by our assumption that $H^0(X_s, \mathcal{O}_{X_s}) = \kappa(s)$. Thus the rank of $\mathcal{E}$ is constant on the fibres. Since $f$ is open (Morphisms, Lemma morphisms-lemma-fppf-open) and closed we conclude that there is a decomposition $S = \coprod S_r$ of $S$ into open and closed subschemes such that $\mathcal{E}$ has constant rank $r$ on the inverse image of $S_r$. Thus we may assume $\mathcal{E}$ has constant rank $r$. We will denote $\mathcal{E}^\vee = \mathcal{H}om(\mathcal{E}, \mathcal{O}_X)$ the dual rank $r$ module.

By cohomology and base change (more precisely by Derived Categories of Schemes, Lemma perfect-lemma-flat-proper-perfect-direct-image-general) we see that $E = Rf_*\mathcal{E}$ is a perfect object of the derived category of $S$ and that its formation commutes with arbitrary change of base. Similarly for $E' = Rf_*\mathcal{E}^\vee$. Since there is never any cohomology in degrees $< 0$, we see that $E$ and $E'$ have (locally) tor-amplitude in $[0, b]$ for some $b$. Observe that for any $g : T \to S$ we have $p_*(q^*\mathcal{E}) = H^0(Lg^*E)$ and $p_*(q^*\mathcal{E}^\vee) = H^0(Lg^*E')$. Let $j : Z \to S$ and $j' : Z' \to S$ be immersions of finite presentation constructed in Derived Categories of Schemes, Lemma perfect-lemma-locally-closed-where-H0-locally-free for $E$ and $E'$ with $a = 0$ and $r = r$; these are roughly speaking characterized by the property that $H^0(Lj^*E)$ and $H^0((j')^*E')$ are finite locally free modules compatible with pullback.

Let $g : T \to S$ be a morphism. If there exists an $\mathcal{N}$ as in the lemma, then, using the projection formula Cohomology, Lemma cohomology-lemma-projection-formula, we see that the modules

$$p_*(q^*\mathcal{E}) \cong p_*(p^*\mathcal{N}) \cong \mathcal{N} \otimes_{\mathcal{O}_T} p_*\mathcal{O}_{X_T} \cong \mathcal{N} \quad \text{and similarly} \quad p_*(q^*\mathcal{E}^\vee) \cong \mathcal{N}^\vee$$

are finite locally free modules of rank $r$ and remain so after any further base change $T' \to T$. Hence in this case $T \to S$ factors through $j$ and through $j'$. Thus we may replace $S$ by $Z \times_S Z'$ and assume that $f_*\mathcal{E}$ and $f_*\mathcal{E}^\vee$ are finite locally free $\mathcal{O}_S$-modules of rank $r$ whose formation commutes with arbitrary change of base (small detail omitted).

In this situation if $g : T \to S$ be a morphism and there exists an $\mathcal{N}$ as in the lemma, then the map (cup product in degree 0)

$$p_*(q^*\mathcal{E}) \otimes_{\mathcal{O}_T} p_*(q^*\mathcal{E}^\vee) \longrightarrow \mathcal{O}_T$$

is a perfect pairing. Conversely, if this cup product map is a perfect pairing, then we see that locally on $T$ we may choose a basis of sections $\sigma_1, \ldots, \sigma_r$ in $p_*(q^*\mathcal{E})$ and $\tau_1, \ldots, \tau_r$ in $p_*(q^*\mathcal{E}^\vee)$ whose products satisfy $\sigma_i\tau_j = \delta_{ij}$. Thinking of $\sigma_i$ as a section of $q^*\mathcal{E}$ on $X_T$ and $\tau_j$ as a section of $q^*\mathcal{E}^\vee$ on $X_T$, we conclude that

$$\sigma_1, \ldots, \sigma_r : \mathcal{O}_{X_T}^{\oplus r} \longrightarrow q^*\mathcal{E}$$

is an isomorphism with inverse given by

$$\tau_1, \ldots, \tau_r : q^*\mathcal{E} \longrightarrow \mathcal{O}_{X_T}^{\oplus r}$$

In other words, we see that $p^*p_*q^*\mathcal{E} \cong q^*\mathcal{E}$. But the condition that the cup product is nondegenerate picks out a retrocompact open subscheme (namely, the locus where a suitable determinant is nonzero) and the proof is complete. $\square$